

M1 Research Proposal

RF-Only Sensing with a 5.9 GHz for V2X scenario using SDR

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Target paper submission: 18 November 2026

Abstract

Investigate a specific gap between 5GHz sidelink for integrated sensing and communication (ISAC) analysis and reproducible roadside sensing measurements i.e localization: whether a stationary, low-power USRP B210 transmitting a 5.9 GHz V2X-inspired waveform can detect the echo of a non-cooperative moving road target while transmitting communication data. The initial system is RF-only: it uses one B210 as the sensing roadside node and a second B210 as an independent communication receiver. Four additional B210s are available, but multi-RSU sensing is further plan. Literature review identifies analytical NR-V2X sidelink sensing, simulation-based resource allocation, distributed cellular ISAC demonstrations, and camera-radar(automotive) fusion as related but distinct approaches. This M1 document states what that review establishes, what remains unverified experimentally, how can measure the model, and how to test the hypothesis before 18 November 2026. Target-detection results will be claimed later this stage.

1 Problem, motivation, and research boundary

The wireless problem being address in this study is not the general possibility of ISAC; communication signals have already been investigated for sensing[1]. Instead, It is being asked: *Can a low-power roadside SDR, designed as a prototype for a future ETSI-based RSU, use a 5.9 GHz OFDM signal to detect a moving object despite interference from its own transmitter and reflections from stationary surroundings? At the same time, how reliably can a separate receiver decode the data carried by that signal?*

Ericsson describes a distributed multi-site cellular sensing network for non-cooperative drones, including detection and tracking over large areas [1]. That establishes industrial interest in RF-only sensing, but it does not report a directly comparable single-B210 roadside experiment with a V2X-like channel. The closest peer-reviewed sidelink study identified is, Decarli *et al.*, analyzes NR-V2X sensing performance and the influence of PHY and resource-access choices, including multi-vehicle interference [2]. These studies motivate a narrower measurement question rather than an unsupported claim that has less studied in sidelink of ISAC.

We have six USRP B210 SDRs available. For the first paper, B210-1 will be the *single stationary sensing RSU*: one transmit path emits OFDM packets, while a simultaneous receive path records reflected IQ. B210-2 will be an *independent communication receiver*, so packet reception is measured over the air instead of inferred from a cable or from the transmitted samples. B210-3 may be assigned to a direct-path reference or cross-check if needed. The remaining devices are spares or resources for later multi-node experiments. Thus “single RSU” refers to *one sensing site*, not a laboratory restriction to one SDR.

The first paper will address moving-target presence, Doppler sign, and radial-velocity estimation. It may report the coarse range only if it is experimentally resolvable after calibration. As for now not claiming camera-based fusion, semantic target classification, precise angle estimation, dense depth, multi-RSU tracking, standard-compliant PC5 operation, or a measured maximum sensing range that has not been established. An existing camera may provide *offline ground truth* for a marked target trajectory; it will never be an input to the sensing algorithm.

2 What we have established from existing work

2.1 A crucial waveform distinction

AS found that ETSI ITS-G5 specifies a 10 MHz OFDM access-layer PHY [4]. LTE/NR-V2X PC5 sidelink is a different, 3GPP-defined PHY, even when it operates in 5.9 GHz ITS spectrum. therefore to choose a *V2X/ITS-G5-parameter-inspired custom OFDM waveform* for the initial B210 implementation. This is a design decision for a feasible prototype, not a claim of ETSI ITS-G5 or NR-V2X PC5 compliance. in final writing it may state the actual occupied bandwidth, subcarrier map, packet structure, channel center frequency, transmit power and duty cycle. There will be no call for a custom packet “PC5” unless a standard-compliant sidelink transmitter/receiver is implemented and verified.

2.2 Focused literature review and SOTA position

Table 1: Focused review: what the approaches establish and what remains different in our setup.

Work	Method and assumptions	Metrics / established result	Limitation relative to our question
Decarli <i>et al.</i> (2024) [2]	Analytical performance characterization of beyond-5G NR-V2X sidelink sensing under PHY/access choices and interfering vehicles.	Detection capability and estimation-quality lower bounds.	Closest <i>sidelink sensing theory</i> reference; not our measured B210 leakage, clutter, or roadside target data.
Li <i>et al.</i> (2025) [3]	Q-learning-assisted resource allocation for full-duplex NR-V2X sidelink JCS; system-level simulation.	Sensing CRLB, packet reception rate, update delay.	Optimizes collisions and resource use; assumes a system model rather than validating one low-power roadside SDR echo.
Ericsson case [1]	ISAC Distributed sensing over multiple cellular sites with analytics for non-cooperative drones.	Public description of network-scale position, speed and tracking capabilities.	A different architecture and target class; public page is not a quantitative benchmark for our hardware.
Singh <i>et al.</i> (2023) [6]; TacoDepth (2025) [7]	RGB camera fused with <i>dedicated automotive radar point clouds</i> for dense depth.	Depth MAE/RMSE; TacoDepth also analyzes model efficiency.	Requires camera input and radar point clouds. Neither is a camera-free 5.9 GHz waveform-echo detector.

Our present SOTA position is therefore *conditional*: Decarli *et al.* is our closest analytical NR-V2X sidelink sensing reference, while Li *et al.* is a relevant communication–sensing resource-allocation reference [2, 3]. We will use them to motivate expected delay/Doppler limitations and communication constraints, not claim to beat their metrics with a different waveform and experimental setup. The camera–radar papers are retained only to explain why existing depth-fusion networks are *not* appropriate baselines for this RF-only experiment [6, 7]. There is a need to repeat a targeted literature search for directly comparable single-site 5.9 GHz SDR roadside experiments before making any final novelty claim.

3 Our signal model and testable hypothesis

In network model, B210-1 as an OFDM transmitter with N active subcarriers, spacing Δf , M OFDM symbols per sensing window, and known transmitted symbols $X[m, n]$. The first-order receive model, after coarse synchronization and FFT, is

$$Y[m, n] = X[m, n] \left(H_{\text{leak}}[m, n] + H_{\text{static}}[m, n] + \alpha e^{-j2\pi n \Delta f \tau} e^{j2\pi m T_{\text{sym}} f_D} \right) + W[m, n], \quad (1)$$

where H_{leak} denotes direct TX–RX leakage, H_{static} stationary clutter, α the target reflection, τ the round-trip delay, f_D target Doppler, and W residual interference/noise. This simplified

single-target model gives us a processing baseline; experiments must reveal whether phase noise, clipping, synchronization and multipath invalidate its clean separation.

With $f_c \simeq 5.9$ GHz, wavelength $\lambda = c/f_c$, and monostatic geometry,

$$R = \frac{c\tau}{2}, \quad v_r = \frac{\lambda f_D}{2}. \quad (2)$$

To estimate the frequency response on known, nonzero transmitted resource elements, $\hat{H}[m, n] = Y[m, n]/X[m, n]$, remove an empty-scene reference, and examine the resulting delay Doppler surface. It need to compare three receivers: raw processing; empty-scene subtraction; and subtraction followed by a fixed-threshold or CFAR-style detector. We will not call a stationary reflection a detected object merely because it appears in a delay map.

The theoretical constraints already explain why moving-target Doppler is our primary task:

$$\Delta R \approx \frac{c}{2B_{\text{occ}}}, \quad \Delta v_r \approx \frac{\lambda}{2T_{\text{coh}}}, \quad (3)$$

where B_{occ} is *occupied*, not nominal channel, bandwidth and T_{coh} is effective coherent time. A 10 MHz *occupied* bandwidth would give approximately 15 m nominal range resolution. That figure is a two target resolution scale, not an automatic lower bound on isolated-target estimation error. In practice, strong zero delay leakage can make close target ranging substantially harder. Coherent integration may sharpen Doppler estimates, but timing drift and target acceleration limit useful integration.

For a monostatic echo the approximate link budget is

$$P_{\text{echo}} \simeq \frac{P_t G_t G_r \lambda^2 \sigma}{(4\pi)^3 R^4 L}, \quad (4)$$

with transmit power P_t , antenna gains G_t, G_r , target radar cross-section σ and lumped loss L . The work have *not* measured these values sufficiently to predict a reliable range yet. Rather than asserting a sensing distance, we will record transmit settings, receiver headroom, directional antenna geometry and detection probability as distance changes.

Testable hypothesis is: *after preventing RX saturation and suppressing repeatable leakage/clutter, a moving high-reflectivity target produces a detectable Doppler component with a controlled false-alarm rate, while an independent B210 receives the OFDM payload.* If repeated trials fail to support the hypothesis, the reported outcome will be the measured operating limit and failure mode, not an inflated target-detection claim.

4 How we are performing the study

4.1 Hardware roles and measurement sequence

The setup uses one B210 at the sensing roadside position and another as a spatially separate packet receiver. B210-1 TX and RX use different, appropriately isolated antennas and a controlled, marked sensing sector. It begin with cabled/attenuated tests to verify sample timing, OFDM demodulation and RX safety, then record target-absent captures to characterize direct leakage and static clutter, followed by repeated moving-reflector or single-vehicle passes at documented positions. B210-2 simultaneously logs correctly decoded packet counts, RSSI/SNR proxies and payload timestamps. B210-3 can supply a reference path if direct leakage subtraction is insufficient; it will not silently become a second sensing RSU. The B210 has two transmit and two receive channels, but simultaneous TX/RX capability alone does not solve self-interference [5].

It may record both positive and negative trials, preserve raw IQ and configuration metadata, and time-align the radio data with a marked-path/video ground-truth log. One trial means a predefined observation window and a predefined yes or no detector decision, avoiding inflated sample counts from strongly correlated adjacent FFT frames. The reported results will include at least one uncertainty interval or trial-count statement for every probability estimate.

4.2 Evaluation criteria

For N_1 target-present windows, N_0 target-absent windows and K velocity-labeled detections, our principal measured quantities are

$$\hat{P}_D = \frac{N_{\text{detected}|\text{target}}}{N_1}, \quad \hat{P}_{FA} = \frac{N_{\text{detected}|\text{empty}}}{N_0}, \quad \text{RMSE}_v = \sqrt{\frac{1}{K} \sum_{k=1}^K (\hat{v}_{r,k} - v_{r,k})^2}. \quad (5)$$

The independent communication receiver allows us to report

$$\text{PER} = \frac{N_{\text{incorrectly decoded packets}}}{N_{\text{transmitted packets}}}, \quad G = \frac{N_{\text{correctly decoded payload bits}}}{T}. \quad (6)$$

A target is declared useful for the paper only when detection improves over the no-target condition at a stated P_{FA} and the radar processing does not assume unavailable image/ground-truth information. We will show raw-versus-canceled delay Doppler maps, ROC or detection-at-fixed-false-alarm results, radial-velocity error, and independent receiver PER vs goodput. We will evaluate an ISAC trade-off *only if* we vary an actual time/frequency/pilot or packet parameter and measure both effects; adding nominal sensing symbols without a matched communications comparison is not sufficient.

4.3 Feasibility limits we have identified

The main technical risk is not just low transmitted power: direct-path leakage may overwhelm echoes or saturate B210-1's receive chain before software cancellation is possible. We therefore treat antenna isolation, gain setting and empty-scene controls as prerequisites. Other risks are inadequate occupied bandwidth for range discrimination, loss of coherent phase across packets, large static clutter, weak/nonrepeatable target echoes and limited trial diversity. We address them with a short, controlled geometry, measured background, repeated passes and explicit reporting of negative results.

5 Timeline and GitHub submission

The target *paper* submission date is 18 November 2026. Detailed plan is given in the Table 2.

The M1 repository will contain The README will give the compilation command. This section documents how the work has being done; it does not claim experiments have already been performed.

References

- [1] Ericsson, "Drone detection with ISAC for defense," accessed 25 September 2026. <https://www.ericsson.com/en/industries/defense/drone-detection-isac>.
- [2] N. Decarli, S. Bartoletti, A. Bazzi, R. A. Stirling-Gallacher, and B. M. Masini, "Performance Characterization of Joint Communication and Sensing With Beyond 5G NR-V2X Sidelink," *IEEE Transactions on Vehicular Technology*, vol. 73, no. 7, pp. 10044–10059, 2024, doi: 10.1109/TVT.2024.3365770.
- [3] Z. Li, P. Wang, Y. Shen, and S. Li, "Reinforcement Learning-Based Resource Allocation Scheme of NR-V2X Sidelink for Joint Communication and Sensing," *Sensors*, vol. 25, no. 2, art. 302, 2025, doi: 10.3390/s25020302.
- [4] ETSI EN 302 663 V1.3.1, "Intelligent Transport Systems (ITS); ITS-G5 Access layer specification for Intelligent Transport Systems operating in the 5 GHz frequency band," 2020. https://www.etsi.org/deliver/etsi_en/302600_302699/302663/01.03.01_60/en_302663v010301p.pdf.

Table 2: Work plan as of 25 September 2026; experimental milestones are not completed results.

Period	What we are doing	Evidence expected
Sep 25–Oct 2	Finalizing waveform-family terminology, relevant literature, mathematical model, permissions, target protocol, and GitHub M1 package.	M1 source; literature matrix with links; block diagram; signed-off scope and safety checklist.
Oct 3–Oct 16	Implementing known-symbol OFDM TX, B210-1 IQ capture, B210-2 packet receiver and cabled timing tests.	Reproducible waveform/code and independent-receiver packet baseline; RX clipping check.
Oct 17–Oct 30	Measuring empty-scene leakage; optimizing antenna isolation; collecting authorized repeated target/no-target passes.	Timestamped IQ, capture settings, ground-truth trajectories and empty-scene controls.
Oct 31–Nov 7	Processing delay–Doppler data; comparing raw and cancelled detectors; calculating detection, false alarm, Doppler error and PER.	Figures with trial counts, error bars/intervals and stated failure cases.
Nov 8–Nov 17	Auditing the novelty claim; writing, revising, and packaging reproducible results.	Complete paper, code/configuration, concise methods and limitations.
Nov 18	Submitting the paper.	Submission record.

- [5] Ettus Research, “USRP B200/B210 Bus Series specification sheet.” https://www.mouser.com/datasheet/2/690/b200_b210_spec_sheet-3236990.pdf.
- [6] A. D. Singh *et al.*, “Depth Estimation from Camera Image and mmWave Radar Point Cloud,” *CVPR*, 2023. Supplied paper.
- [7] Y. Wang *et al.*, “TacoDepth: Towards Efficient Radar-Camera Depth Estimation with One-stage Fusion,” *CVPR*, 2025. Supplied paper.